Online Resource 1

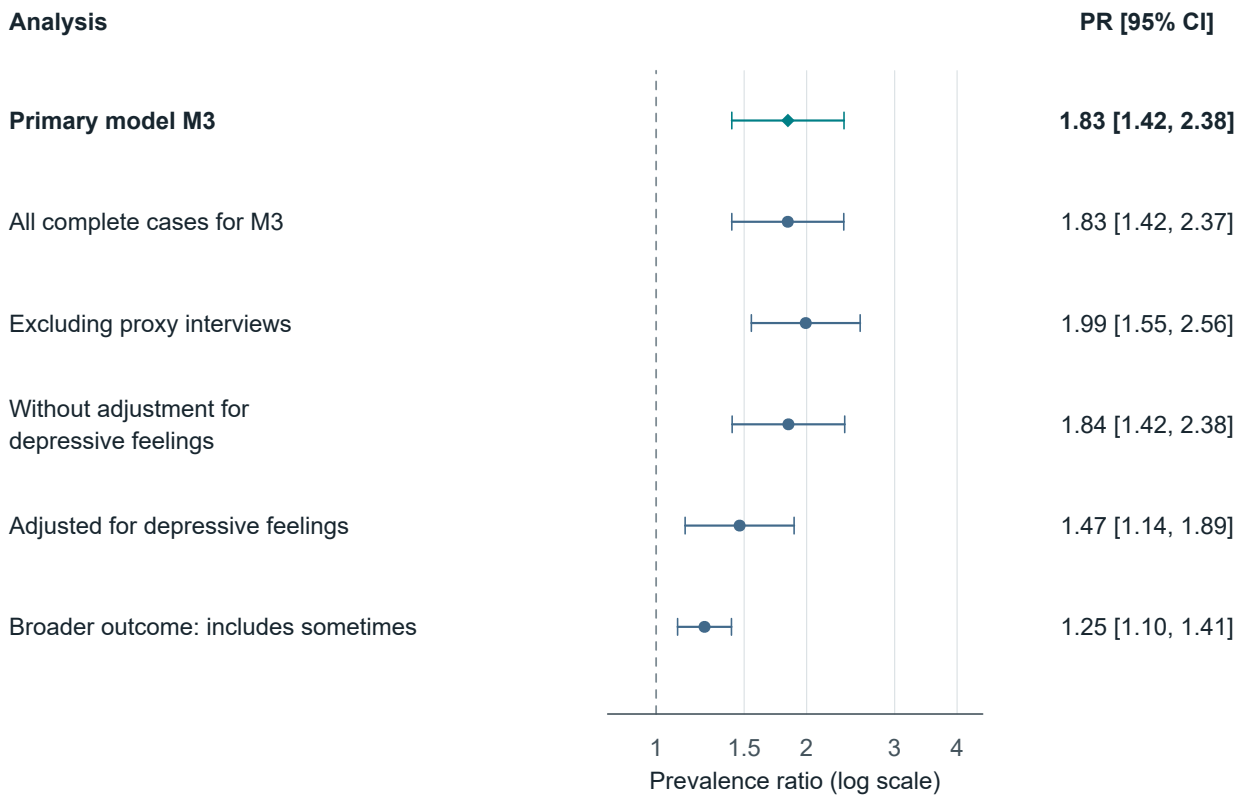


Fig. S1 Sensitivity analyses of the association between swallowing problems and loneliness, NHIS 2025

Points and lines indicate prevalence ratios (PRs) and 95% confidence intervals. The primary M3 model is included for comparison. Additional rows show M3 in all complete cases for that model, M3 excluding proxy interviews, models without and with additional adjustment for depressive feelings in an identical sample (N=8,055), and M3 with a broader loneliness outcome. Depressive feelings were classified as daily/weekly versus less often/never; this variable does not indicate a clinical depression diagnosis. The broader outcome includes always, usually or sometimes feeling lonely (2,120 events). Comparisons must account for differences in sample composition and outcome definition. Supplementary analyses are not adjusted for multiplicity. Some fitted values exceed 1 in the depressive-feelings-adjusted and broader-outcome Poisson models; these fits should not be interpreted as individual probabilities. Attenuation after adjustment for depressive feelings does not establish causal mediation. The logistic sensitivity analysis is not plotted on the same axis because its effect measure is an odds ratio, not a PR.

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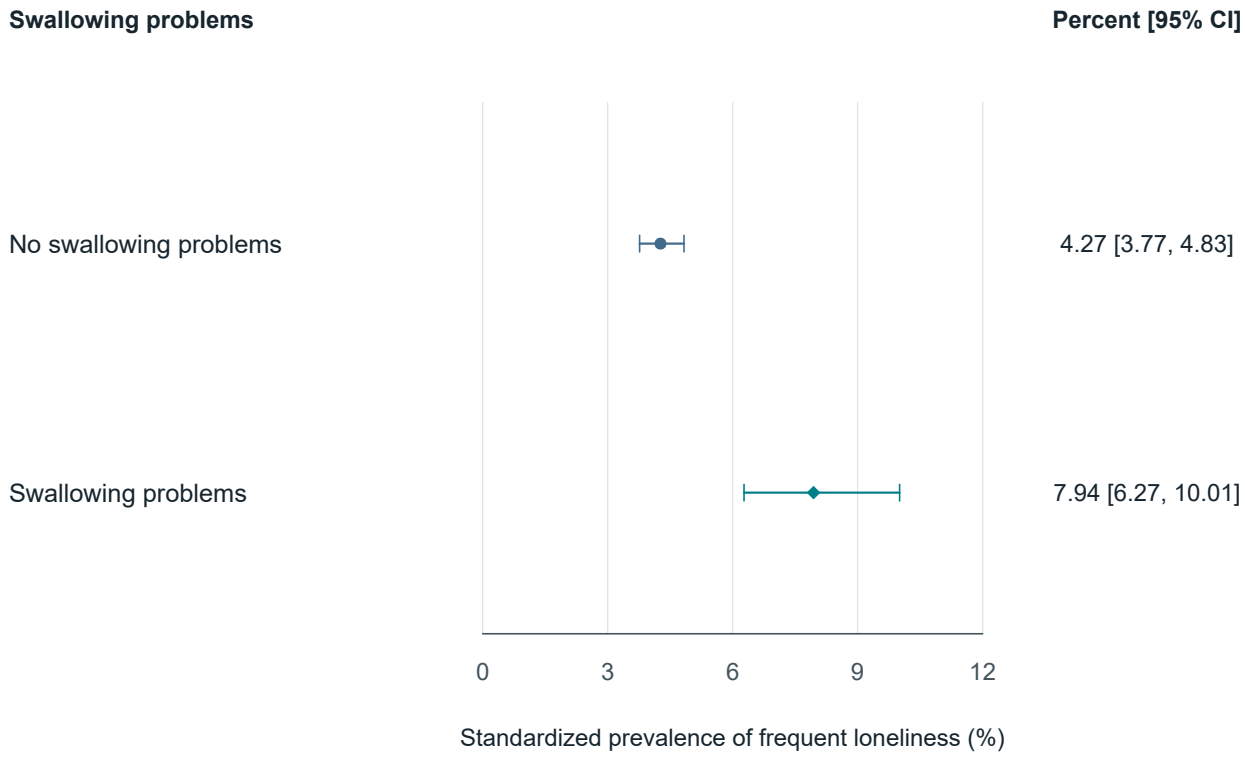


Fig. S2 Model-standardized prevalence of frequent loneliness by swallowing problems among adults aged 65 years or older, NHIS 2025

Points and horizontal lines show standardized prevalences and 95% confidence intervals from survey-weighted logistic models with the M3 covariates. Frequent loneliness was defined as always or usually feeling lonely. Swallowing problems refer to self-reported difficulties during the preceding 12 months. Both exposure levels were standardized over the survey-weighted covariate distribution of the same complete-case sample of 8,094 adults, including 429 outcome events. Covariates were age group, sex, education, family income-to-poverty ratio, partnership, living alone, mobility, hearing, stroke history, in-person contact and telephone/video contact. Estimates and variances were pooled over ten official income imputations. Prevalences were pooled on the logit scale, the absolute difference on its natural scale and the marginal prevalence ratio on the log scale, so the pooled contrasts need not exactly equal arithmetic operations on the separately rounded pooled prevalences. These model-standardized associations are not causal effects. The marginal PR is not the same parameter as the primary conditional Poisson PR. Related estimates: absolute prevalence difference 3.67 percentage points (95% CI 1.75, 5.60); marginal prevalence ratio 1.86 (95% CI 1.43, 2.42).

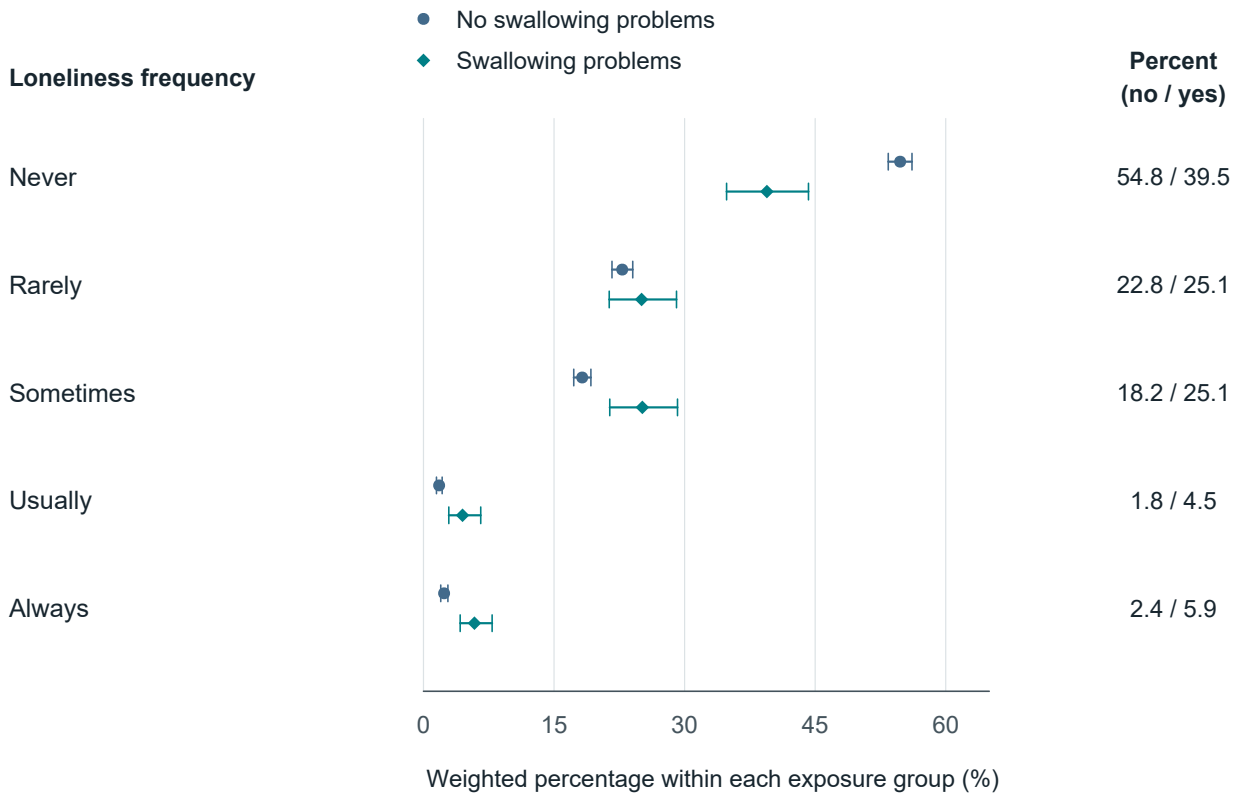


Fig. S3 Survey-weighted distribution of all five loneliness response categories by swallowing problems among adults aged 65 years or older, NHIS 2025

Points represent weighted category percentages and horizontal lines show Korn-Graubard 95% confidence intervals. Blue circles indicate participants without swallowing problems; teal diamonds indicate participants with swallowing problems. The analysis includes 7,545 and 686 participants, respectively, with valid exposure and outcome responses; the two groups contain 8,231 adults in total. Percentages sum to 100% within each exposure group before rounding. The sample sizes are unweighted. These distributions are unadjusted for covariates; all ten category proportions satisfy the implemented NCHS precision criteria. The primary binary outcome combines the usually and always categories. No common proportional-odds estimate is superimposed because the planned diagnostic did not support the full common-slope specification. The distributions and uncertainty intervals describe cross-sectional self-report data and do not establish direction or causation.